Dynamic Individuals, Static Neighborhoods: Migration, Earnings Changes, and Concentrated Poverty*

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Abstract

We use administrative data to document a high degree of migration across neighborhoods and neighborhood types defined in terms of poverty rate and median income. Neighborhood quality increases over an individual's life cycle, and people also move to better neighborhoods in response to earnings improvements. We then develop several implications of these initial facts for high-poverty neighborhoods. First, resident turnover in these areas is rapid. Second, among the people living in a poor neighborhood at a point in time, the distribution of future concentrated poverty exposure is bimodal. Young people, renters, and those with children tend to spend fewer than half of the next ten years in similar neighborhoods, while older people and homeowners are unlikely to exit concentrated poverty. Third, poor neighborhoods tend to remain poor because of a dynamic process in which initial residents experience high earnings growth but disproportionately out-migrate when earnings improve, contrasting with a pure "poverty trap" understanding of persistent concentrated poverty.

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1 Introduction

Concentrated poverty and income segregation are defining features of neighborhoods in the United States. A growing body of research finds that exposure to high-poverty neighborhoods has negative long-run consequences for employment and educational outcomes, as well as physical and mental health (Kling et al., 2007; Chetty et al., 2016; Chyn, 2018). This has generated debate over the appropriate policy response. On one hand, policy could reduce exposure to disadvantaged neighborhoods by offering mobility assistance, as in the Moving to Opportunity program. On the other hand, neighborhood revitalization programs, such as Empowerment Zones or Opportunity Zones, attempt to improve conditions in the worst-off neighborhoods for those who stay put (Busso et al., 2013; Freedman et al., 2021; Gaubert et al., 2021).

Our understanding of poor neighborhoods and these policy approaches is limited by the scarcity of longitudinal data on the migration and economic trajectories of the people who live in them. How common is migration into and out of different types of neighborhoods, and what factors drive moves? Are individuals in poor neighborhoods at a point in time likely to remain in similar areas and experience sustained low earnings, or do they see changes over time? How do these individual-level patterns relate to neighborhood-level trends, such as the high persistence of concentrated poverty?

In this paper, we address these questions using newly available administrative data that contain an annual panel of neighborhood locations for most U.S. residents. We connect these data to American Community Survey (ACS) responses and the Longitudinal Employer-Household Dynamics (LEHD) data to provide a comprehensive picture of individual demographics, labor market outcomes, and location choices. We begin by documenting some new basic facts about migration across neighborhoods and its relationship with earnings dynamics. We then use these facts as the basis for an improved understanding of both individuals' exposure to concentrated poverty and the mechanisms underlying the persistence of poor neighborhoods. The results have implications for the design and relative merits of people- and place-based policy.

We first provide novel evidence on the degree to which people migrate across different types of neighborhoods over time. We pay particular attention to high-poverty neighborhoods, defined as census tracts with a poverty rate over 30%. This is roughly the poorest decile of tracts, and the resulting set of neighborhoods is similar to Empowerment Zones in size and demographics. Individual mobility is high, with 45% of adults changing census tracts over a 10-year period, rising to 50% among people initially living in high-poverty neighborhoods. This migration is not just between similar areas. Over a quarter of people initially in high-poverty neighborhoods live in a neighborhood with poverty below 20% eight years later, with half of the migrants landing in a neighborhood below 10% poverty. Accordingly, the set of people living in a place turns over quickly, even when considering areas larger than a tract. Roughly 45% of adults in high-poverty areas at time t have moved to a tract below our high-poverty threshold, left the CBSA, or died by time $t + 10$. These initial migration results build directly on a sociology literature that uses data from the Panel Survey on Income Dynamics (PSID) to compute similar statistics on migration across types of neighborhoods (Gramlich et al., 1992; Massey et al., 1994; South and Crowder, 1997; Quillian, 2003; Brummet and Reed, 2021).[1]

Next, we consider two explanations for the large amount of migration across neighborhoods. First, we show that cross-neighborhood mobility has an important life cycle component: individuals live in poorer neighborhoods when young and move to better neighborhoods as they age, perhaps due to either increasing resources or changing priorities at different life stages. A quarter of 25 year-olds live in a tract with poverty above 20 percent, falling to 17 percent at age 55. The share living in low-poverty tracts commensurately increases by nearly 10 percentage points. However, life cycle changes in neighborhood conditions are considerably smaller than the aggregate across-neighborhood migration rates above, which highlights an important role for other factors.

Second, we find that individuals' neighborhood choices are significantly impacted by idiosyncratic changes in their earnings, even along the life cycle. We begin by documenting a strong

[1]We build on this literature by using a much larger data set that allows measurement over longer time periods, within much finer subgroups of people and neighborhoods, and with more precision. These papers find similar results to ours where the estimated quantities coincide, providing some cross-validation of data sources.

correlation between earnings changes and neighborhood mobility. Among individuals beginning in a high-poverty tract, those whose annual earnings increase by $20,000 from t to $t+8$ are 13 percentage points more likely to have exited concentrated poverty in $t+8$ than those who experience no earnings growth, and they end up in tracts with $7,200 higher median income.

We then estimate the causal effect of idiosyncratic earnings shocks. To do so, we use the design in Rose and Shem-Tov (2023) to examine the effects of idiosyncratic, firm-level pay shocks, which we infer using changes in coworker earnings following Koustas (2018) and Ganong et al. (2020). We find that a shock that increases annual household earnings by 1.37 log points on average over an eight-year period leads to increases in tract median incomes and housing values of 0.24 and 0.29 log points, respectively, implying elasticities of 17.5 percent and 21.2 percent. These results show that while individuals do not perfectly re-sort after earning changes, shocks to earnings do causally generate churn across neighborhood types.[2]

The importance of the strong relationship between earnings and neighborhood choices depends on the frequency and size of earnings changes. Prior literature has documented large earnings volatility generally and especially rapid growth among people with low earnings at baseline (Abowd et al., 2018; Guvenen et al., 2021). However, it has not conditioned on an individuals' neighborhood, which may affect labor market outcomes through channels such as commuting access and job referral networks (Bayer et al., 2008). We leverage our unique data to verify that these general patterns occur within different types of neighborhoods. After conditioning on individual characteristics, people living in high-poverty neighborhoods at baseline have very similar future earnings growth as people in less poor areas, including a long right tail of large earnings increases. This suggests that the selective migration we documented is quantitatively important in understanding poor neighborhoods, and it also adds another dimension of volatility or churn. Individuals are not only frequently moving across neighborhoods—the characteristics (like earnings) that we use to describe them are also changing over time.

Overall, our individual-level results show that people in poor neighborhoods experience a sub-

[2]This exercise complements prior work by Bilal and Rossi-Hansberg (2021), who study movement across regions in response to mass layoffs, and Golosov et al. (2023), who study the migration and housing choices of lottery winners.

stantial amount of volatility in both neighborhood choices and earnings, and individual earnings are an important determinant of this migration. In the final part of the paper, we study how incorporating churn and the correlation between earnings and migration can improve our understanding of exposure to concentrated poverty and its effects on individuals, as well as the mechanisms that perpetuate high neighborhood poverty.

We leverage the panel data to compare longitudinal measures of concentrated poverty exposure to cross-sectional measures. To get a sense of the difference between longitudinal and cross-sectional exposure metrics generally, we follow the approach in Kopczuk et al. (2010) and compute measures of neighborhood inequality using each. We find that longitudinal measures are 7 to 15 percent lower than cross-sectional, suggesting that people observed in the tails of the neighborhood quality distribution at a point in time tend to subsequently move towards the center. We then zoom in on people living in high-poverty tracts at a point in time and find a bimodal distribution of future concentrated poverty exposure. About 54 percent will be in a high-poverty place for all of the next ten years, while about 31 percent spend five or fewer of those years in concentrated poverty. However, there are sharp differences between demographic groups. Among those under age 35, a third spend all ten years in a high poverty neighborhood, while half spend five or fewer years there. Similarly, young people with children are more likely to spend fewer years in a high poverty neighborhood, which is of interest because the negative effects of concentrated poverty are largest among young children (Chetty et al., 2016).[3]

Finally, we illustrate how the individual-level patterns we have documented thus far jointly result in highly persistent neighborhood poverty.[4] To do so, we first compute growth in mean earnings among the cohort of prime-age adults who lived in different neighborhood types in 2005, regardless of where they lived in later years. Growth is quite large over the subsequent ten years for every neighborhood type (between 24 and 26 percent in real terms), consistent with our earnings

[3]Quillian (2003) studies concentrated poverty exposure with the PSID and reaches similar aggregate conclusions. As with the earlier literature on migration across neighborhoods, the scale and detail of the administrative data allow us extend his work in a number of ways.

[4]Malone and Redfearn (2018) show that U.S. census tracts have not changed much since 1970. Rosenthal and Ross (2015) show that neighborhoods changed frequently over a longer time period that spans upheavals such as mass suburbanization and the Great Migration.

dynamics results. We then compute the mean earnings in each year for a different group—the contemporaneous residents of each tract type. In low-poverty tracts, contemporaneous earnings grows at about the same speed as that of the baseline residents. That is, the mean earnings of low-poverty tract residents in 2015 is similar to the 2015 mean earnings of people who lived in low-poverty tracts in 2005. This is not true in high poverty neighborhoods, where contemporaneous earnings grows by only 10 percent. The divergence arises because individuals sort across neighborhoods when their earnings change. This sorting process prevents mean earnings in high-poverty neighborhoods from improving at the same rate as those of their baseline residents, which will tend to keep poor neighborhoods poor even if labor market fluctuations or policy interventions boost individual outcomes.

This paper contributes to three related literatures. First is work on neighborhood change, broadly defined to include specific types of change (e.g., Card et al. 2008; Couture and Handbury 2020), neighborhood persistence (e.g., Rosenthal 2008; Lee and Lin 2018; Malone and Redfearn 2018), and evaluations of neighborhood revitalization policy (e.g., Busso et al. 2013; Freedman et al. 2021). Due to data limitations, these studies generally cannot observe the individual mechanisms that drive neighborhood-level trends.[5] Our results suggest that these individual-level processes are likely quite dynamic. For example, concentrated poverty appears to persist because, although many baseline residents see significant earnings increases, they disproportionately move away when they do. As a result, improving economic outcomes of initial residents might have limited effects on a neighborhood, because those residents quickly diffuse due to high baseline migration rates, and, moreover, become more likely to exit if their earnings improve.

In addition, a large body of work estimates the causal effects of neighborhoods on individual outcomes, typically focusing on poor neighborhoods and exploiting variation produced by housing safety net programs (Kling et al., 2007; Chetty et al., 2016; Chyn, 2018). Our paper complements this literature by using a population-level sample to provide comprehensive estimates of the migration patterns that underlie exposure to different types of neighborhoods. Most notably, we find

[5]Recent work showing that gentrification primarily occurs through changes in in-migration rather than out-migration is a notable exception (McKinnish et al., 2010; Ellen and O'Regan, 2011; Brummet and Reed, 2021).

that concentrated poverty exposure is transitory for a substantial share of individuals, but much longer-lasting for others. Neighborhood effect estimates could differ between those subgroups, potentially reconciling differences in the literature. More generally, our results emphasize that the estimated effects of the Moving to Opportunity program are relative to a control group that exhibits high mobility and significant neighborhood improvement (Kling et al., 2007).

Finally, the most directly related research also measures across-neighborhood migration and earnings dynamics. This literature has largely relied on the PSID to follow individuals over time. Access to administrative data allows us to build on prior work in numerous ways, as discussed above. In addition, we connect individual-level statistics to their neighborhood implications in more detail than prior work.

2 Data and Sample Construction

2.1 Data Sources

We draw longitudinal address location information from the Census Bureau's Master Address File-Auxiliary Reference File (MAFARF). The MAFARF includes the geolocated housing unit of residence for the near-universe of adults in the U.S. in each year from 2000 to 2021, which is derived from federal tax, health, and housing records.[6] We use these data to identify each individual's (2010 vintage) census tract in each year, which we use as our definition of neighborhood throughout the paper.

Earnings data is taken from the LEHD data. Our LEHD extract contains 25 states and at least the 2000 to 2021 time period for all states. The data includes quarterly employment and earnings for individuals whose employers participate in unemployment insurance, as well as employer information, such as industry NAICS code.[7] The included states are representative of the country in terms of population, region, and demographic composition.[8] We adjust earnings (and all other

[6]Graham et al. (2017) and Voorheis et al. (2023) discuss the administrative data sources used to construct the MAFARF, and Sullivan and Genadek (2024) describe how the data can be used for research on migration.

[7]Abowd et al. (2018) provide a useful summary of the LEHD's construction and coverage rate.

[8]Our LEHD extract consists of Arizona, California, Colorado, Connecticut, Delaware, Indiana, Iowa, Maine,

dollar-valued variables used in the analysis) to 2019 dollars using the Consumer Price Index.

Individual and household demographics are drawn from the ACS, which is an annual survey spanning roughly three percent of households in the U.S. For each individual, we observe demographics including age, sex, education, and race/ethnicity. At the household level, we observe whether a household rents or owns their primary residence, their estimated house value or monthly rent, and total household income. In addition, the ACS contains self-reported measures of earnings, wages, and hours worked that supplement the LEHD data.

Finally, we construct neighborhood characteristics by aggregating ACS responses to the tract level. We calculate poverty rates, median household incomes, and median home values using the 2005 to 2009 waves of ACS, which coincides with the start of our sample period. We use time-invariant measures of neighborhood characteristics to ensure that these variables are not influenced themselves by the migration that we study.

2.2 Sample Construction

We combine these data sources to create a large and representative panel of individuals containing information on demographics, residential location, and labor market outcomes. We then use that overarching panel to construct a migration sample, which we use for exercises that do not require earnings information, and an earnings sample.

The initial sampling frame consists of respondents to the 2005 to 2013 waves of the ACS. We restrict to individuals who, at the time they were sampled in the ACS, were at least 25 years old, not full-time students or in the armed forces, not the child of the household head, and not living in group quarters or a census tract in which more than 20 percent of the population is between the ages of 18 and 25. The last restriction, which affects about one percent of census tracts, is intended to remove neighborhoods with high amounts of college student housing from the sample.

We define the baseline year for each individual as the year that they were sampled in the ACS.[9]

Maryland, Massachusetts, Nevada, New Jersey, New Mexico, North Dakota, Ohio, Oklahoma, Pennsylvania, South Carolina, South Dakota, Tennessee, Utah, Virginia, Washington, Wisconsin, and Wyoming.

[9] In the event that an individual was sampled twice, we keep only their first response.

We then merge in longitudinal address information from the MAFARF for the three years prior to and ten years following the baseline. For people living in a state included in our LEHD extract in the baseline year, we then add information on employment and earnings.

Throughout the paper, we group tracts into broad categories based on their poverty rate. The categories are 0 to 10 (roughly the bottom half of tract poverty rates), 10 to 20 (percentiles 50 to 75), 20 to 30 (percentiles 75 to 90), and 30 to 100 percent (the highest decile). The highest poverty category has similar characteristics to the distressed tracts targeted in neighborhood revitalization policies such as Empowerment Zones. For robustness, we also use population-weighted quintiles of median household income (defined nationally) to define neighborhood categories in some exercises.

Depending on the exercise, we use different subsamples of this overarching panel. Defining these samples separately highlights that we measure migration with no requirements on labor market attachment, while our earnings dynamics results require some restrictions on earnings and age that are common to the literature.

Migration sample: For results on migration that do not incorporate any earnings or employment information, we use the matched ACS-MAFARF data for the full U.S. In order to include ten years of neighborhood locations following the baseline ACS survey, we include only respondents to the 2005 to 2011 ACS waves. The migration sample ultimately consists of 13.2 million unique individuals, including 670,000 who live in the high-poverty category in the baseline year.

Earnings dynamics sample: For results on earnings dynamics or the relationship between earnings and migration, we use the matched ACS-MAFARF-LEHD data. We include the 2005 to 2013 ACS waves and use an eight-year follow-up window.[10] In addition to reducing the number of states included, we further restrict the sample to include only people with at least moderate labor force attachment. We drop individuals who were over age 55 or reported working fewer than 1042 hours in the ACS. The resulting sample contains 1.7 million unique individuals, including 60,000 in high-poverty tracts at baseline. Finally, to prevent outliers from driving our estimates, we also treat

[10] We make this slight change from the selection criteria of the migration sample to reduce the share of observations that begin during the peak of the Great Recession.

person-years in which an individual was observed to earn less than $4,000 or over $300,000 in the LEHD as missing. Similar sample restriction are common in the literature on earnings dynamics (e.g. Abowd et al. 2018).

3 Rates of Migration Across Neighborhoods

We begin by showing that people move across census tracts quite frequently, especially in relatively poor neighborhoods. Figure 1 plots the probability that an individual lives in their baseline year tract in each of the ten subsequent years. We report separate statistics for people who begin in different types of neighborhoods, with Panel A classifying neighborhoods by poverty rate and Panel B by quintile of median income. Overall, we find that about 57 percent of people live in the same tract ten years later, or, equivalently, that 43 percent of people have changed tracts. Panel A shows that rates of out-migration are elevated for higher poverty neighborhoods: where the poverty rate exceeds 30 percent, 51 percent of individuals live in a different neighborhood 10 years later. Panel B shows patterns that are qualitatively similar when classifying neighborhoods based on their median income.[11]

The economic significance of these moves depends on the types of neighborhoods people are moving to. For example, if individuals from poor neighborhoods simply move to other poor neighborhoods, there may be little improvement in the conditions or opportunities they face. To explore this issue, we report the full matrix of transitions across different neighborhood types in Table 1, using an eight-year period to capture a medium-run perspective. There is a large amount of mass off the diagonal, especially among people originating in poorer neighborhoods. For example, 26.6 percent of people who begin in the highest poverty category transition to a tract with poverty below twenty percent eight years later, including 12.7 percent to the lowest poverty category.[12] The

[11] Brummet and Reed (2021) find that about half of the residents of initially low-income, central city tracts in 2000 had moved by the time they were observed in 2010–2014. Despite the difference in sample and data sources, this is quite close to our estimate for low-income or high-poverty neighborhoods. Because we focus on a broader set of geographies and a later and expanded set of years, the similarity of these two estimates suggests a fair degree of recent stability in out-migration rates from poor neighborhoods.

[12] For comparison, the one-year out-migration rate is 6.1 percent (Appendix Table A.1). Extrapolating this one-year out-migration rate forward as a Markovian transition probability would imply an eight-year out-migration rate

small relative size of the highest-poverty category (roughly 10 percent of tracts and 5 percent of individuals in the sample) mechanically increases its out-migration rate, but we similarly see that 17.8 percent of the bottom tract income quintile transitions to the third quintile or above by year 8. There are also flows in the negative direction, although they are somewhat less common.[13] Among people originating in the middle income quintile, 12.6 percent have moved down eight years later, while 16.8 percent have moved up. The numbers are more skewed in Panel A due to the size differences between the categories, with 20 percent of the 10 to 20 poverty group transitioning up, versus only 6.6 percent transitioning down.

These high rates of individual migration have an important implication: neighborhoods turn over quickly. Because census tracts are generally smaller than both popular conceptions of neighborhoods and the areas targeted by revitalization policies, we conclude this section by calculating turnover at a much broader scale than in Figure 1. We use our poverty and income neighborhood categories above and calculate the "stayer rate" as the share of baseline residents of a neighborhood category who remain in the same category and CBSA.[14] Figure 2 shows that the stayer rate in the high-poverty category has fallen to 50 percent by ten years after the baseline, while it is nearly 70 percent in the lowest poverty category. The difference is due to both individual mobility rates and the difference in the size of each type. These large estimates are likely lower bounds on the true rate of neighborhood turnover given the large size of our categories, and we consider their implications for neighborhood revitalization policy in the final section.

Similar statistics on transition rates between neighborhood types appear in a number of papers that take advantage of the PSID's addition of respondents' tract location in the 1990s (Gramlich et al., 1992; Massey et al., 1994; South and Crowder, 1997). This work similarly finds high migration rates across high- and low-poverty areas, but the small size of the PSID limits them to one-year transitions and coarser neighborhood categories. When the estimated quantities coincide, we find

of 39.6 percent ($= 1 - (1 - 0.061)^8$), which is much higher than what we estimate directly. This is consistent with heterogeneous migration propensities leading to dynamic self-selection Bayer and Juessen (2012).

[13] This is consistent with both net population declines in poor neighborhoods and individuals migrating to less poor neighborhoods as they age.

[14] By this definition, people who are not "stayers" have (1) moved to a different category within the same CBSA, (2) moved to a different CBSA, or (3) died.

similar results to these papers, which suggests that across-neighborhood migration rates have not changed much over time.

One important implication from these results is that a person's location in a single cross-section may not reflect their longitudinal neighborhood exposure. Instead, individuals appear to reside in considerably different types of neighborhoods throughout their lives. These findings motivate the rest of the analyses in this paper.

4 Determinants of Migration Across Neighborhoods

In this section, we provide evidence on the role of two key determinants of these evolving location choices. First, we study life cycle patterns, which could lead individuals to move to better neighborhoods as they age due to either increasing resources or changing tastes. Second, we consider the role of changes in earnings. We provide a combination of descriptive and causal estimates that highlight important roles played by both of these factors.

4.1 Life Cycle Patterns

The top panel of Figure 3 shows that migration rates vary greatly by age. Nearly 30 percent of 25 and 26 year-olds move in a given year, versus 15 percent of 35 year-olds and only 6 percent of 55 year-olds. For all age groups, between 50 and 60 percent of moves are across tracts within a county. This steep gradient in overall migration rates likely reflects a number of ways that moving costs increase with age, including increasing homeownership rates and growing household sizes. It may also occur because people gradually discover and move to their desired location. Regardless of the reason, it is clear that a disproportionate share of migration across neighborhoods is driven by young people.

The bottom panel of Figure 3 shows how individuals' experienced neighborhood characteristics tend to improve over the course of the life cycle.[15] The share of individuals who reside in a high poverty neighborhood falls from 11 percent at age 25 to 7 percent at age 64. This represents a 36

[15] As in Section 3, we measure neighborhood conditions at a single point in time to isolate the role of migration.

percent decrease in the share of people living in a high poverty neighborhood over four decades; by comparison, the results in Table 1, Panel B show a comparable movement out of high poverty neighborhoods over just an eight-year period. We view this evidence as suggesting that life cycle forces can explain some, but likely not all, of the cross-neighborhood migration patterns, which leads us to focus on the role of changes in earnings next.

4.2 Earnings Changes and Neighborhood Choice

Along a life cycle trajectory, neighborhood choices might diverge across individuals as they experience idiosyncratic shocks. In particular, unexpected shocks to earnings (such as a layoff or new job opportunity) are a potentially important driver of neighborhood mobility. In this subsection, we examine how neighborhood choice responds to changes in earnings.

4.2.1 Descriptive Evidence

We first examine the relationship between earnings changes and neighborhood choice descriptively. Panel A of Figure 4 plots the eight-year change in the 2005–2009 median income of an individual's census tract against the change in their annual earnings.[16] Because the relationship may differ for individuals beginning in different types of neighborhoods, we stratify based on individuals' baseline tract poverty rate. We find that people with larger increases in earnings systematically move to higher-income tracts, with a stronger relationship among individuals who begin in higher-poverty neighborhoods. In the highest poverty category, a $10,000 increase in annual earnings is associated with a $2,500 larger increase in tract income. In contrast, among individuals beginning in the lowest poverty neighborhoods, a $10,000 increase in earnings is associated with only a $200 increase in tract income.[17]

[16] Recall that results on earnings use a more restricted sample than the migration results, including requirements that that individuals work at least half-time in their ACS years and are under age 56.

[17] Some related work has studied the correlation between changes in individual poverty status and migration. Using Canadian administrative data, Frenette et al. (2004) find that less than half of migration into or out of concentrated poverty coincides with a large earnings change or change in household composition. Using the PSID, Massey et al. (1994) do not find that people who moved in a year are more likely to have entered or exited poverty in the same year.

The strong correlation between earnings growth and migration across neighborhoods illustrates how persistent concentrated poverty might coexist with high earnings growth among baseline residents—people with good income realizations could simply move away. To illustrate this dynamic directly, Panel B of Figure 4 examines the relationship between changes in earnings and exit from the highest-poverty tracts for people originating in these tracts. The baseline probability that an individual in the earnings sample exits the set of high-poverty neighborhoods is about 44 percent.[18] This increases rapidly as the earnings change increases, reaching 57 and 74 percent for earnings changes between \$20,000 to \$30,000 and over \$60,000, respectively. We explore this dynamic from a neighborhood perspective in Section 6.

4.2.2 Quasi-Experimental Methodology

While these correlations are suggestive, it is possible that they reflect expected life cycle earnings growth rather than responses to unforeseen shocks. We next study the causal impacts of unforeseen earnings changes on neighborhood choice by exploiting firm-wide pay shocks at one's initial employer, as in Koustas (2018) and Ganong et al. (2020). Isolating the role of earnings changes is valuable for distinguishing among the mechanisms that determine migration and for considering place-based policies.

The idea is to compare observationally similar individuals who are employed at observationally similar firms at baseline, but whose firms' payroll grows at different rates. Following Rose and Shem-Tov (2023), we focus on individuals who are observed in the ACS in some year between 2005 and 2013 and can be matched to the LEHD. We further restrict to workers who report working at least 1040 hours in the ACS (roughly half-time employment) and for whom we can identify at least 25 coworkers in the LEHD data.

We construct idiosyncratic firm-level changes in average pay for each individual using a "hold-out" sample that consists of all of the individual's coworkers who were observed in the LEHD but

[18] This rate of exit is slightly higher than the exit rate of 38 percent in Table 1. Individuals in the earnings sample are more attached to the labor market and younger than individuals in the migration sample, which helps explain the higher rate of exit.

not sampled in the ACS during the 2005 to 2013 period.[19] We define each individual's pay shock as the mean percent earnings change in their hold-out sample, measured from the baseline quarter to four quarters later.[20] Using year-over-year changes in earnings limits the impact of seasonality, and our restriction to individuals who have at least 25 LEHD coworkers limits the noise in the shock variable

To compare observably similar individuals at similar firms, we include both individual-level and firm-level controls in our primary specification. The individual controls include: initial wage, 2005–2009 median income of the tract of residence in the ACS, age, sex, race, ethnicity, and education. Firm-level controls (computed using the hold-out sample) include: the log of firm employment, mean pay, median pay, average separation rates, average new worker accession rates, and average separations into non-employment averaged over the four quarters prior to the baseline quarter. Finally, we also include fixed effects to control for differential trends across industries and local labor markets.

The resulting regression regression specification is:

$$Y_{i,t,k} = \mu_i + \sum_{k=-5,k\neq-1}^{8} \beta_k \text{PayShock}_{i,t} + \theta_{k\times t_q} + \omega_{n(i)\times s(i)\times t_y\times k} + \lambda_{c(i)\times t_y\times k} + X_i'\alpha_{t,k} + \epsilon_{i,t,k} \quad (1)$$

where $Y_{i,t,k}$ is the outcome of interest, such as log earnings or log tract income, for individual i in year k relative to initial ACS response time t. The pay shock for person i, which is defined as of time t, is $\text{PayShock}_{i,t}$. We allow the coefficients on this variable to vary with relative time, k, and scale it so that coefficients can be interpreted as the effects of a 10 percent year-over-year increase in coworker earnings. We control for common changes over time using the fixed effects $\theta_{t_q\times k}$, which capture the interaction between the ACS quarter of observation (t_q) and relative year of observation (k). The fixed effects $\omega_{n(i)\times s(i)\times t_y\times k}$ allow trends in outcomes to vary arbitrarily

[19] We require these workers to earn more than $2,600 at baseline (roughly half-time employment at minimum wage) and be employed at the same firm from the quarter before baseline (i.e., the quarter when a person is observed in the ACS) until the fifth quarter after baseline. These are thus coworkers who do not separate from their firm. Since our main sample is drawn from ACS respondents, this holdout sample and the main sample do not overlap.

[20] To limit the influence of outliers, we successively winsorize individual earnings, the percent change in individual earnings, and the pay shock at the 99th percentile.

based on individuals' baseline 2-digit NAICS industry codes ($n(i)$), baseline state of residence ($s(i)$), and ACS year of observation (t_y). The fixed effects $\lambda_{c(i)\times t_y \times k}$ additionally absorb variation in changes over time by baseline CBSA of residence ($c(i)$) and ACS year of observation. X_i is the vector of individual and firm-level controls, which are interacted with relative time and time of ACS response. We cluster our standard errors at the firm level.

4.2.3 Quasi-Experimental Results

We plot the dynamic effects of the coworker pay shocks on individual earnings in Panel A of Figure 5. An increase in coworker earnings of 10 percent is associated with an immediate increase in own annual earnings of about 2.5 percent. Although the shocks are only defined based on changes in coworker earnings in a single year, their effect on annual earnings reverts only gradually, remaining elevated by 1 percent in year 8. Accordingly, *cumulative* earnings gains/losses grow over time.

While we define the coefficients as the effects of a net positive shock, the shock variable includes both positive and negative changes and impacts workers through multiple margins. We find that a net positive shock increases the probability of remaining at one's initial firm, but the relative magnitudes of the effects on earnings and job exit imply that shocks increase earnings within job spells as well. Since households might smooth income in response to shocks through adjustments to partners' labor supply, and total household income is most relevant for relocation decisions, we also display effects on *household* earnings, defined as combined LEHD earnings of individuals and their partners (as reported in their ACS response).[21] While the impact on individual earnings is partially offset by partner earnings, the combined effect on household earnings is only slightly smaller than the effect on individual earnings.

Turning to the effects on migration, Panel B of Figure 5 shows that individuals with more positive earnings shocks are more likely to move, with an effect size peaking at 0.4 percentage points by the final year. Our earlier results suggest that the baseline probability of moving over an eight-year period is roughly 40 percentage points, implying a 1 percent increase in the probability

[21] For individuals who do not report a partner in the ACS, individual and household income are the same.

of moving in response to a persistent 1–2 percent increase in annual earnings. Interestingly, the effect on the probability of moving to a higher-income tract than one's baseline, shown in the green markers, is nearly the exact same magnitude. This suggests that almost all induced relocation is towards higher-income neighborhoods.[22]

Figure 6 considers the relationship between the earnings shock and measures of neighborhood quality. More positive shocks are correlated with moves towards tracts with higher 2005–2009 median income and median housing values. In contrast to the effects on earnings, which fade over time, the effects on neighborhood quality *increase* over time, reaching roughly 0.25 percent at their peak.[23] One useful approach to assessing the magnitude of these responses is to benchmark the end-year effect on tract income (0.24 log points) or home values (0.29 log points) relative to the average effect on household earnings over the post-period (1.37 log points), which reflects the persistent change in earnings rates. This yields elasticities of 0.175 and 0.212, respectively. Thus, while individuals do not perfectly re-sort after income shocks, they do systematically move to better neighborhoods.

4.3 Earnings Dynamics in Different Neighborhoods

This relationship between earnings changes and neighborhood choice suggests that individual earnings volatility could drive substantial cross-neighborhood mobility. Prior literature has shown that earnings are volatile generally, and growth is especially rapid among people with low earnings at baseline (Abowd et al., 2018; Guvenen et al., 2021). Similarly, there is significant churn into and out of poverty, resulting from changes in both income and household composition (Bane and Ellwood, 1986; Stevens, 1999). However, little is known about how these earnings dynamics vary across people living in different neighborhoods. A key question is whether individuals in

[22] While the increase in short-run mobility after earnings shocks is consistent with the increase in move rates among lottery winners documented by Golosov et al. (2023), we find that moves after persistent earning shocks are more directed to higher-income neighborhoods than moves after one-off lottery winnings.

[23] One potential explanation of these dynamics is that lumpy relocation decisions are irreversible, and are therefore made based on expectations of long-run income rather than contemporaneous income, such that mobility increases as the persistence of income shocks becomes evident. Relocation decisions may also be sensitive to *accumulated* earnings gains or losses, particularly for credit-constrained individuals buying homes.

high-poverty neighborhoods experience large enough earnings growth and volatility to explain the churn that we observe in those areas.

In this subsection, we characterize earnings growth in different types of neighborhoods.[24] Table 2 shows results from a series of regressions of individual earnings changes (measured from the baseline year to eight years later) on neighborhood and individual characteristics. The results in Columns 1 and 2 of Panel A indicate the average eight-year earnings increase in dollars is higher in lower-poverty neighborhoods, though the results in Column 3 show that those differences substantially reflect differences in the characteristics of people living in different neighborhoods. However, Panel B shows that average growth in *percentage* terms is highly similar across all neighborhood types, whether or not one controls for individual characteristics. On the whole, these regressions suggest that differences across neighborhoods in earnings growth are largely explained by the types of individuals that live in the neighborhood, rather than by the neighborhood itself. While we do not have a causal estimate of the effects of neighborhoods on earnings growth, this is consistent with past work that found that neighborhoods do not have a large effect on adult labor market outcomes (Kling et al., 2007). An interesting implication is that there is significant earnings growth even among low-income individuals originating in poor neighborhoods.

Next, we go beyond the average and plot the distribution of earnings changes in different neighborhood types. Figure 7 shows the distribution of eight-year earnings changes (in dollars) among people originating in each tract poverty category. Lower poverty neighborhoods have a relatively high number of both large negative changes in earnings (possibly reflecting labor force exits) and large positive changes. Moreover, a large number of people in poor neighborhoods see significant labor market improvements, likely large enough to motivate some to change neighborhoods.

[24]Our analysis extends earlier work by Massey et al. (1994), who used the PSID to compute the probability that individuals living in different neighborhoods transition into or out of poverty in the subsequent year.

5 Implications for Measuring Exposure to Neighborhood Conditions

An extensive literature studies racial segregation and inequality in neighborhood conditions. Recent work showing that neighborhood conditions are an important determinant of individuals' outcomes has drawn increased attention to this topic (Chyn and Katz, 2021, e.g.). Our finding of a substantial amount of migration across different types of neighborhoods raises the possibility that cross-sectional measures of exposure to neighborhood conditions could misstate individuals' longer-term experience. In this section, we show that there is less inequality in neighborhood conditions and exposure to concentrated poverty when individuals' neighborhood exposure is measured over a longer time period.

5.1 Inequality in Neighborhood Conditions

We start by computing some very general statistics on neighborhood inequality using both longitudinal and cross-sectional measures of neighborhood exposure. We define cross-sectional exposure using an individual's census tract in the year they were sampled in the ACS, and we define longitudinal exposure as the mean characteristics of the census tracts an individual lived in during a ten-year interval centered on the ACS year. To facilitate comparisons with measures of earnings inequality, we use the earnings subsample for this exercise, and we further restrict to people over age 29 when sampled in the ACS in order to accommodate the five-year look-back component of the longitudinal measure.

Specifically, we follow the approach in Shorrocks (1978) and Kopczuk et al. (2010) in comparing short-term neighborhood conditions, denoted for person i in year t as y_{it}, and long-term neighborhood conditions, denoted as $\bar{y}_i \equiv \sum_{t=-5}^{5} y_{it}/10$. Letting $\text{var}_i(\bar{y}_i)$ denote the variance over i of long-term neighborhood conditions and $\sum_{t=-4}^{5} \text{var}_{it}(y_{it})/10$ denote the average of the short-term variance measures, we can construct the Shorrocks mobility index, $M \in [0, 1]$ as:

$$M = 1 - \frac{\text{var}_i(\bar{y}_i)}{\sum_{t=-4}^{5} \text{var}_{it}(y_{it})/10}. \tag{2}$$

We focus on the log of tract median income as our measure of neighborhood conditions for this exercise. The variance in cross-sectional log tract income is 0.159, versus a variance of 0.137 for average log tract income. These numbers imply a Shorrocks mobility index for neighborhood conditions of $M_{\text{nhood}} = 1 - 0.137/0.159 = 0.138$. In other words, the estimated variance of long-term neighborhood conditions is almost 14 percent lower than the estimated average variance of short-term conditions. By way of comparison, when focusing on log earnings of the same individuals, we estimate a Shorrocks mobility index of $M_{\text{earn}} = 1 - 0.316/0.359 = 0.120$.[25] This implies that there is slightly more mobility in terms of neighborhood conditions than in terms of earnings. As another way of quantifying this phenomenon, we find that the spread between the 90th and 10th percentile of tract income is 7.6 percent lower using the average over a ten-year period ($69,100) versus a cross-sectional measure ($74,810).

These substantial reductions in inequality measures occur because people observed in the tails of the neighborhood distribution at one point in time revert towards the mean in the longitudinal measures. In short, people change neighborhoods frequently enough that mean reversion from point-in-time measures is substantively important.

5.2 Exposure to Concentrated Poverty

The difference between longitudinal and cross-sectional measures could be particularly significant for research on high-poverty neighborhoods. These areas are in the tail of the distribution, where mean reversion tends to be strongest in general. Moreover, people who are likely to live in high-poverty neighborhoods have high migration rates and volatile earnings, likely leading them to have high year-to-year variance in their neighborhood quality.

We next compute some more detailed statistics on the longitudinal neighborhood experience of people who live in concentrated poverty at a point in time. To do so, we take the set of people living in tracts with poverty rates over 30 percent in their ACS survey year and calculate the share of the following ten years that they spend in similarly poor neighborhoods. We freeze tract poverty

[25]This is quite similar to the estimated Shorrocks mobility index of 0.135 for year 2002 from Kopczuk et al. (2010) (Table II).

rates at their 2005–2009 ACS levels and consider neighborhood change separately.

Table 3 shows the distribution of concentrated poverty exposure for the full subsample and a set of subgroups. Exposure is generally bimodal. Overall, 31 percent of individuals observed in a high-poverty neighborhood at baseline spend no more than 5 out of the next 10 years in these types of tracts, while 54 percent are in concentrated poverty for all ten years. Column 5 shows that exits from concentrated poverty are typically to significantly different neighborhoods—the average person who has exited concentrated poverty by year five lives in a tract with poverty rate 24 percent below their baseline tract.

The lower rows of the table show that exposure varies widely across demographic subgroups. The share of people spending all of the next 10 years in a high poverty neighborhood is much higher among older individuals: only 31 percent of people who are 25–35 years old at baseline spend all 10 years in a high poverty neighborhood, compared to 70 percent of people over age 55. People who are renters in the baseline year also spend less subsequent time in high poverty neighborhoods relative to those who are homeowners. Exposure to concentrated poverty is more persistent for Black and Hispanic individuals than for White individuals, though the racial difference is smaller in magnitude than the differences by age and homeownership. People who have children living in their household at the time of the ACS spend fewer years in a high poverty tract, driven by people who are 40 or under at the time of the ACS.

Finally, we consider neighborhood change in columns 6 and 7. We start with the group that spends ten years in a tract with a high level of poverty from 2005–2009. As calculated in column 6, nearly 25 percent ($= 0.127/.538$) of this group lives in a tract in year 10 where the 2015–2019 poverty rate is below 25 percent, which represents a drop of at least five percentage points from the tract's poverty rate in the 2005–2009 ACS. On average, the poverty rate falls by 15 percent for the individuals in column 6 who see this type of improvement. This illustrates that a significant share of people also exit concentrated poverty through neighborhood change. Notably, the share of people who exit concentrated poverty through neighborhood change is largest for the groups who are least likely to move out of their initial neighborhood.

This exercise is closely related to Quillian (2003), who uses the PSID to consider exposure to concentrated poverty. Our findings are consistent with their PSID results—for example, we both find that out-migrants reduce their tract poverty by about 20 percentage points. However, the administrative data allows us to expand the analysis significantly. We can consider finer subgroups, identifying the sharp age gradient and differences between households with and without children, which are important features for connecting to the neighborhood effects literature. We also have a large enough sample to define a higher tract poverty threshold (30 versus 20 percent), which is more similar to the neighborhoods typically targeted by revitalization policies.[26] Although it is focused on a different outcome, prior work on poverty spells (i.e. the amount of time that an individual is poor) also has a similar flavor. Bane and Ellwood (1986) and Stevens (1999) find a bimodal distribution in which many people experience only short spells in poverty, but a large mass is also persistently poor.

These results also have some interesting connections to the neighborhood effects literature. In particular, they highlight that the estimated treatment effects of programs like Moving to Opportunity (MTO) that encourage movement out of high-poverty areas are relative to a control group that exhibits very high baseline outbound mobility and improvement in neighborhood conditions. In the original MTO studies, the control group saw their neighborhood poverty rate decrease from 50 percent at the time of random assignment to 31 percent 10 to 15 years later (Kling et al., 2007). While one might suppose this baseline mobility was an artifact of focusing on a sample of individuals who had volunteered for a program that encouraged moves, we show that similar rates of mobility are actually typical of the broader population of concentrated poverty residents, particularly among young residents and those with children. At the same time, we also find a large mass of people who stay in high poverty neighborhoods for long periods of time, indicating important heterogeneity in propensities to move neighborhoods. Treatment effects could differ between these subgroups, due to differences in either household characteristics or counterfactual neighborhood

[26]Some related work considers how the probability of exiting high-poverty areas varies with household characteristics and years spent in high-poverty areas (South and Crowder, 1997; Frenette et al., 2004). Similarly, Lee et al. (2017) and Sampson and Sharkey (2008) study specific sequences of neighborhood locations.

exposure. This may help reconcile differences in the literature.[27]

6 Implications for Neighborhood-Level Change

There is a large amount of research on neighborhood change (e.g., Cutler et al., 1999; Lee and Lin, 2018; Baum-Snow and Hartley, 2020; Couture and Handbury, 2020; Bartik and Mast, 2022). However, most relevant to our study is a pair of papers that document the strong persistence of neighborhood conditions since 1970 (Sampson and Morenoff, 2006; Malone and Redfearn, 2018).

We have already shown in Section 3 that there is a substantial amount of migration across different types of neighborhoods. This suggests that the evolution of neighborhood-level outcomes occurs through large gross flows of individuals. In this section, we add to prior work by illustrating the individual migration and earnings dynamics that underlie the persistence of neighborhood composition during our sample period. This expands our knowledge of the mechanisms of neighborhood persistence and provides some general features of poor neighborhoods that may inform the design of place-based policies.[28]

We first compute growth in mean earnings among the cohort of people who lived in each tract poverty category in the 2005 ACS, regardless of where they lived in later years (restricting to the 1965 to 1980 birth cohorts to minimize the roles of retirement and labor force entry.) As shown in Figure 8, earnings growth is quite large over the subsequent ten years for every category, between 24 and 26 percent in real terms. This is consistent with our earnings dynamics results, again illustrating the point that people across all types of neighborhoods see earnings growth on average.

For comparison, we next compute the mean earnings in each year for a different group—the contemporaneous residents of each tract type. In low-poverty tracts, contemporaneous earnings

[27]For example, the contrast between the finding in Chyn (2018) that individuals move to better neighborhoods after public housing demolitions and the more modest effects found in studies of voluntary voucher programs might be explained by the fact that demolitions induced moves among individuals with both high and low relocation propensities alike.

[28]Several recent papers have taken a similar perspective to gentrification and shown that changes in in-migration patterns, rather than involuntary out-migration of incumbent residents, is the primary mechanism through which this type of neighborhood change occurs (McKinnish et al., 2010; Ellen and O'Regan, 2011; Brummet and Reed, 2021; French et al., 2023). Sampson and Sharkey (2008) and Quillian (1999) also connect individual changes to neighborhood outcomes using a flow approach.

grow at about the same speed as the earnings of baseline residents. That is, the mean earnings of low-poverty tract residents in 2015 is similar to the 2015 mean earnings of people who lived in low-poverty tracts in 2005. This is not true in either the 20–30 or 30+ percent poverty neighborhoods, where contemporaneous earnings grows by only 11 percent over the ten-year period. This wedge appears because of another fact that we documented at the individual level—people whose earnings increase tend to move to better neighborhoods.

This comparison illustrates the dynamic process underlying the persistence of concentrated poverty. Poor neighborhoods remain poor despite healthy earnings growth among initial residents, because people who experience increases in earnings disproportionately leave and are replaced by poorer in-migrants. This dynamic helps explain why concentrated poverty tends to persist through macroeconomic trends and local economic shocks—selective migration prevents these shocks from passing through to neighborhood-level outcomes. This perspective is distinct from a pure poverty trap story, in which poor neighborhoods would feature depressed earnings growth and low migration rates. Of course, the two stories are by no means mutually exclusive; negative neighborhood effects and the selective out-migration of people with good income realizations can coexist and reinforce each other.

These general features of poor neighborhoods also have some implications for policies that target distressed neighborhoods. Specifically, our results on earnings growth suggest that neighborhoods could be revitalized in part by retaining the initial residents who have good income realizations. This may be both easier and more politically palatable than improving a neighborhood by attracting residents without a connection to the area. An example policy is Michigan's Renaissance Zone program, which exempts residents from essentially all state and local taxes as long as they live in the zone. Alternatively, retention could be improved through programs that directly target quality of life, such as Community Development Block Grants or the block grant component of Empowerment Zones.

In addition, the high turnover rate of initial residents has implications for policies that try to improve the labor market outcomes of residents of distressed neighborhoods. For example, Empowerment Zones provide tax credits for employers who hire zone residents, and Promise Zones utilize a number of Department of Labor programs. These policies typically target a small set of tracts, roughly ten percent of a metropolitan area, which is on the order of our highest poverty category. Our results suggest that the set of people who are exposed to an intervention at a point in time will subsequently dissipate widely, with most living outside the targeted area within 10 years. Thus, even if the policy successfully improves outcomes of a cohort of residents, the composition of the targeted neighborhood may not change. This is a general challenge for revitalization policies that may help explain these programs' inconsistent and small effects (Neumark and Simpson, 2015). In addition, note that our earlier results on the causal effect of earnings increases on migration implies that the more effectively a policy improves individual outcomes, the larger this problem becomes.

References

Abowd, J. M., McKinney, K. L., and Zhao, N. L. (2018). Earnings inequality and mobility trends in the united states: Nationally representative estimates from longitudinally linked employer-employee data. *Journal of Labor Economics*, 36(S1):S183–S300.

Bane, M. J. and Ellwood, D. T. (1986). Slipping into and out of poverty: The dynamics of spells. *Journal of Human Resources*, 21(1):1–23.

Bartik, A. and Mast, E. (2022). Black suburbanization: Causes and consequences of a transformation of american cities. *Available at SSRN 3955835*.

Baum-Snow, N. and Hartley, D. (2020). Accounting for central neighborhood change, 1980-2010. *Journal of Urban Economics*, 117(103228).

Bayer, C. and Juessen, F. (2012). On the dynamics of interstate migration: Migration costs and self-selection. *Review of Economic Dynamics*, 15(3):377–401.

Bayer, P., Ross, S. L., and Topa, G. (2008). Place of work and place of residence: Informal hiring networks and labor market outcomes. *Journal of political Economy*, 116(6):1150–1196.

Bilal, A. and Rossi-Hansberg, E. (2021). Location as an asset. *Econometrica*, 89(5):2459–2495.

Brummet, Q. and Reed, D. (2021). The effects of gentrification on incumbent residents. Working Paper.

Busso, M., Gregory, J., and Kline, P. (2013). Assessing the incidence and efficiency of a prominent place based policy. *American Economic Review*, 103(2):897–947.

Card, D., Mas, A., and Rothstein, J. (2008). Tipping and the dynamics of segregation. *Quarterly Journal of Economics*, 123(1):177–218.

Chetty, R., Hendren, N., and Katz, L. F. (2016). The effects of exposure to better neighborhoods on children: New evidence from the Moving to Opportunity Experiment. *American Economic Review*, 106(4):855–902.

Chyn, E. (2018). Moved to opportunity: The long-run effects of public housing demolition on children. *American Economic Review*, 108(10):3028–56.

Chyn, E. and Katz, L. F. (2021). Neighborhoods matter: Assessing the evidence for place effects. NBER Working Paper 28953.

Couture, V. and Handbury, J. (2020). Urban revival in America. *Journal of Urban Economics*, 119(103267).

Cutler, D. M., Glaeser, E. L., and Vigdor, J. L. (1999). The rise and decline of the american ghetto. *Journal of political economy*, 107(3):455–506.

Ellen, I. G. and O'Regan, K. M. (2011). How low income neighborhoods change: Entry, exit, and enhancement. *Regional Science and Urban Economics*, 41(2):89–97.

Freedman, M., Khanna, S., and Neumark, D. (2021). Jue insight: The impacts of opportunity zones on zone residents. *Journal of Urban Economics*, page 103407.

French, R., Gandhi, A., and Gilbert, V. (2023). Quantifying the welfare effects of gentrification on incumbent low-income renters.

Frenette, M., Picot, G., and Sceviour, R. (2004). When do they leave? the dynamics of living in low-income neighbourhoods. *Journal of Urban Economics*, 56(3):484–504.

Ganong, P., Jones, D., Noel, P. J., Greig, F. E., Farrell, D., and Wheat, C. (2020). Wealth, race, and consumption smoothing of typical income shocks. Technical report, National Bureau of Economic Research.

Gaubert, C., Kline, P. M., and Yagan, D. (2021). Place-based redistribution. Technical report, National Bureau of Economic Research.

Golosov, M., Graber, M., Mogstad, M., and Novgorodsky, D. (2023). How americans respond to idiosyncratic and exogenous changes in household wealth and unearned income*. *The Quarterly Journal of Economics*, 139(2):1321–1395.

Graham, M. R., Kutzbach, M. J., and Sandler, D. H. (2017). Developing a residence candidate file for use with employer-employee matched data. *Center for Economic Studies Working Paper*, (17-40).

Gramlich, E., Laren, D., and Sealand, N. (1992). Moving into and out of poor urban areas. *Journal of Policy Analysis and Management*, 11(2):273–287.

Guvenen, F., Karahan, F., Ozkan, S., and Song, J. (2021). What do data on millions of us workers reveal about lifecycle earnings dynamics? *Econometrica*, 89(5):2303–2339.

Kling, J. R., Liebman, J. B., and Katz, L. F. (2007). Experimental analysis of neighborhood effects. *Econometrica*, 75(1):83–119.

Kopczuk, W., Saez, E., and Song, J. (2010). Earnings inequality and mobility in the united states: Evidence from social security data since 1937. *The Quarterly Journal of Economics*, 125(1):91–

128.

Koustas, D. (2018). Consumption insurance and multiple jobs: Evidence from rideshare drivers.

Lee, K. O., Smith, R., and Galster, G. (2017). Neighborhood trajectories of low-income us households: An application of sequence analysis. *Journal of Urban Affairs*, 39(3):335–357.

Lee, S. and Lin, J. (2018). Natural amenities, neighborhood dynamics, and persistence in the spatial distribution of income. *Review of Economic Studies*, 85(1):663–694.

Malone, T. and Redfearn, C. L. (2018). Shocks & ossification: The durable hierarchy of neighborhoods in us metropolitan areas from 1970 to 2010. *Regional Science and Urban Economics*, 69:94–121.

Massey, D. S., Gross, A. B., and Shibuya, K. (1994). Migration, segregation, and the geographic concentration of poverty. *American sociological review*, pages 425–445.

McKinnish, T., Walsh, R., and White, T. K. (2010). Who gentrifies low-income neighborhoods? *Journal of Urban Economics*, 67(2):180–193.

Neumark, D. and Simpson, H. (2015). Place-based policies. In *Handbook of regional and urban economics*, volume 5, pages 1197–1287. Elsevier.

Quillian, L. (1999). Migration patterns and the growth of high-poverty neighborhoods, 1970–1990. *American Journal of Sociology*, 105(1):1–37.

Quillian, L. (2003). How long are exposures to poor neighborhoods? the long-term dynamics of entry and exit from poor neighborhoods. *Population research and policy review*, 22(3):221–249.

Rose, E. K. and Shem-Tov, Y. (2023). How replaceable is a low-wage job?

Rosenthal, S. and Ross, S. (2015). Change and persistence in the economic status of neighborhoods and cities. *The Handbook of Regional and Urban Economics*.

Rosenthal, S. S. (2008). Old homes, externalities, and poor neighborhoods. a model of urban decline and renewal. *Journal of urban Economics*, 63(3):816–840.

Sampson, R. J. and Morenoff, J. D. (2006). Durable inequality. *Poverty traps. Princeton University Press, Princeton, New Jersey, USA*, pages 176–203.

Sampson, R. J. and Sharkey, P. (2008). Neighborhood selection and the social reproduction of concentrated racial inequality. *Demography*, 45(1):1–29.

Shorrocks, A. F. (1978). Income inequality and income mobility. *Journal of Economic Theory*, 19:376–393.

South, S. J. and Crowder, K. D. (1997). Escaping distressed neighborhoods: Individual, community, and metropolitan influences. *American Journal of Sociology*, 102(4):1040–1084.

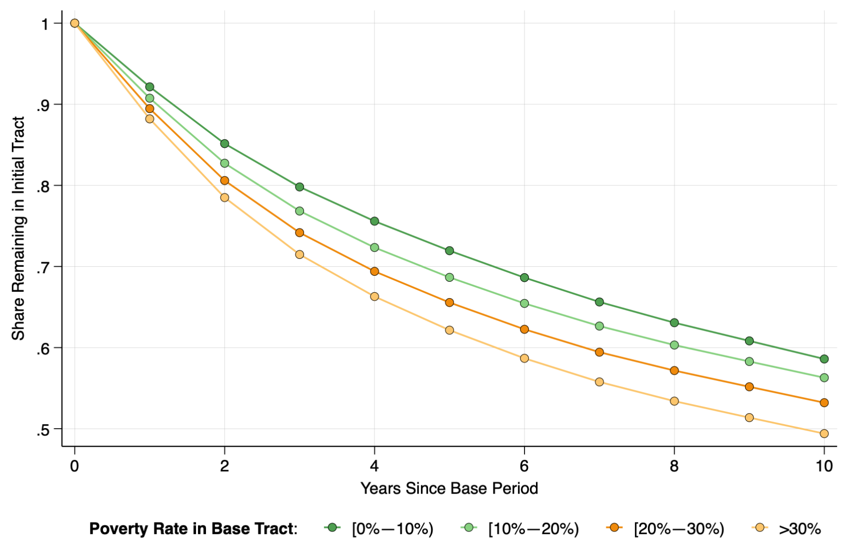
Stevens, A. H. (1999). Climbing out of poverty, falling back in: measuring the persistence of poverty over multiple spells. *Journal of Human Resources*, 34(3):557–560.

Sullivan, J. and Genadek, K. (2024). Using the census bureau's master address file for migration research. Working Paper.

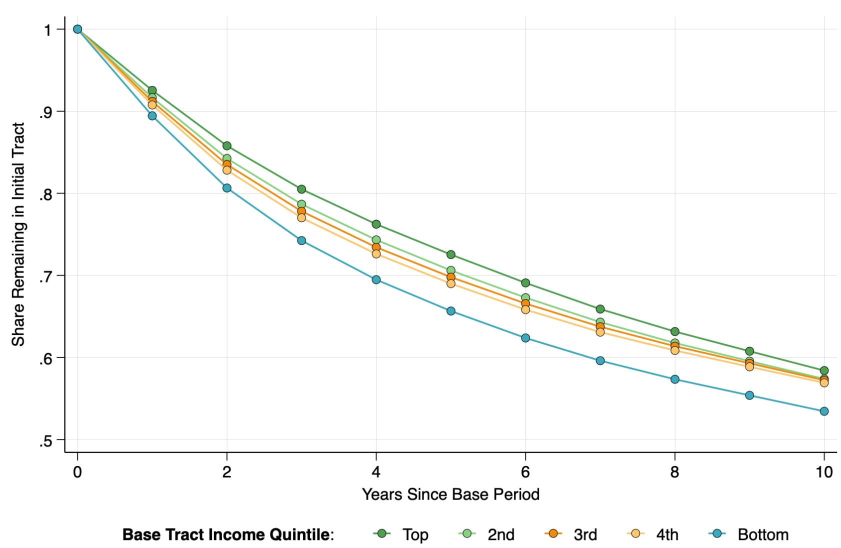
Voorheis, J. L., Colmer, J. M., Houghton, K. A., Lyubich, E., Munro, M., Scalera, C., and Withrow, J. R. (2023). Building the prototype census environmental impacts frame. Technical report, National Bureau of Economic Research.

Figure 1: Share of Individuals Remaining in Baseline Tract Over 10 Years

(a) By Baseline Tract Poverty



(b) By Baseline Tract Earnings

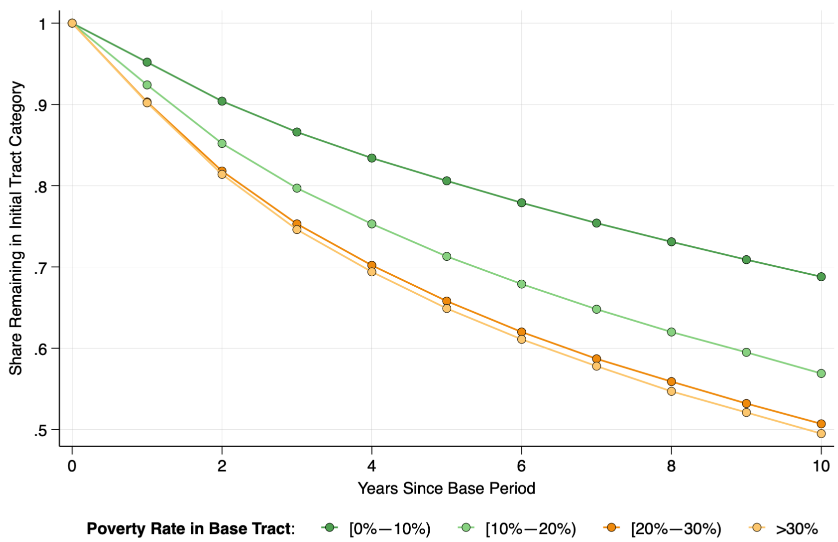


Notes: This figure shows the share of individuals who live in their baseline census tract t years after the baseline year. Individuals are stratified based on the poverty rate of their baseline tract (Panel A) or its income quintile (Panel B). This figure uses the migration sample, as described in Section 2.2, and the baseline year is defined as the year an individual was sampled in the ACS.

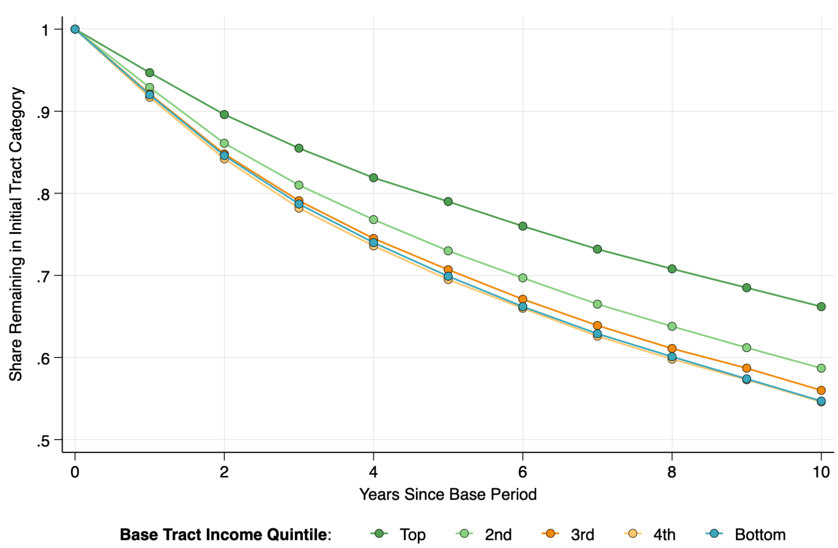
Source: Authors' calculations using data from the ACS and MAFARF.

Figure 2: Share of Individuals Remaining in Baseline Tract *Category* and CBSA Over 10 Years

(a) By Baseline Tract Poverty



(b) By Baseline Tract Earnings

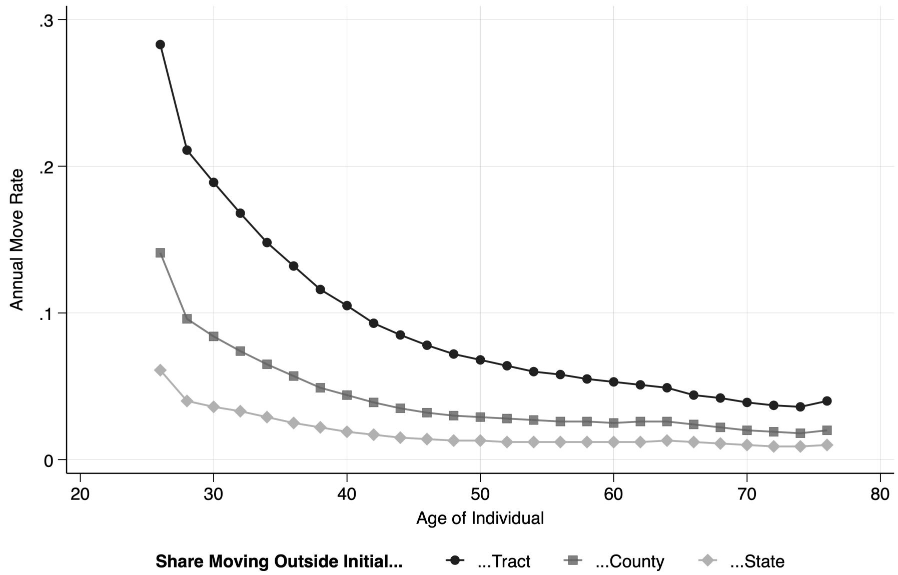


Notes: This figure shows the share of people who remain in the same tract type and same CBSA in which they lived in the baseline year, counting death as an exit. Tract types are defined as the categories listed in each legend. We separately plot these shares for individuals living in tract types defined according to poverty rates (Panel A) or median income levels (Panel B) in the baseline year. The figure uses the migration sample, as described in Section 2.2, and the baseline year is defined as the year an individual was sampled in the ACS.

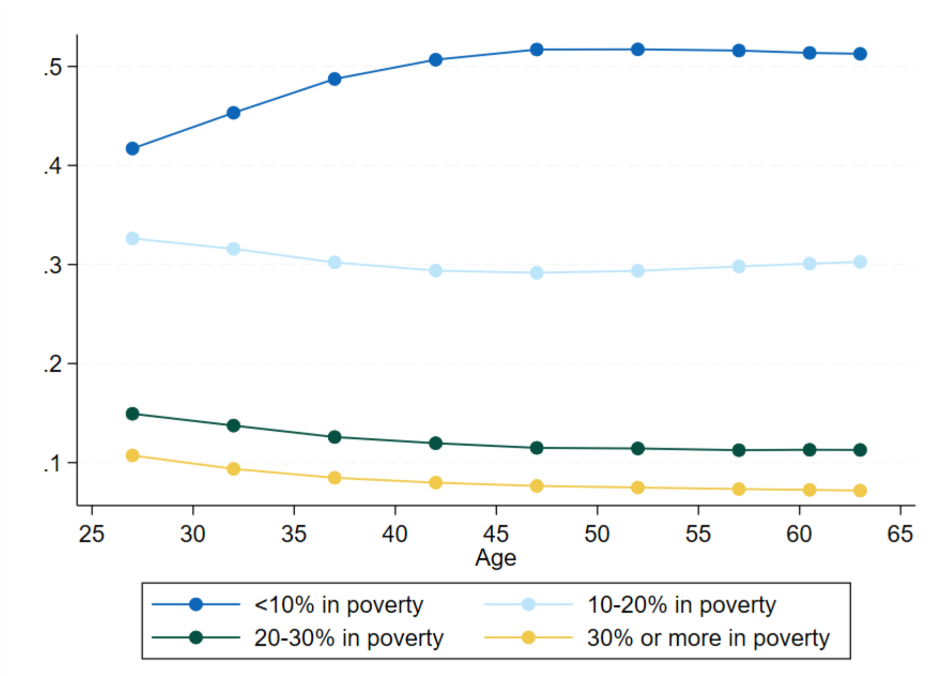
Source: Authors' calculations using data from the ACS and MAFARF.

Figure 3: Annual Neighborhood Mobility, By Age

(a) Migration Rate



(b) Share Living in Tracts With Different Poverty Rates

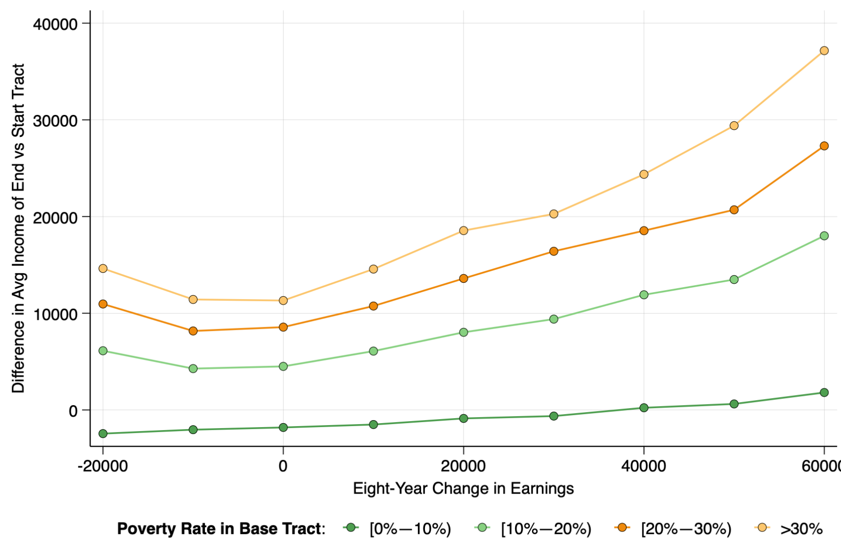


Notes: Panel A displays the share of individuals in each two-year age bin who move across census tract, county, or state lines in a year. This figure uses the migration sample, as described in Section 2.2, and migration is measured over the year after an individual is sampled in the ACS. Panel B displays the average of the log median household income level from 2005–2009 that is experienced by individuals of each age. This comes from publicly available ACS data for all U.S. residents.

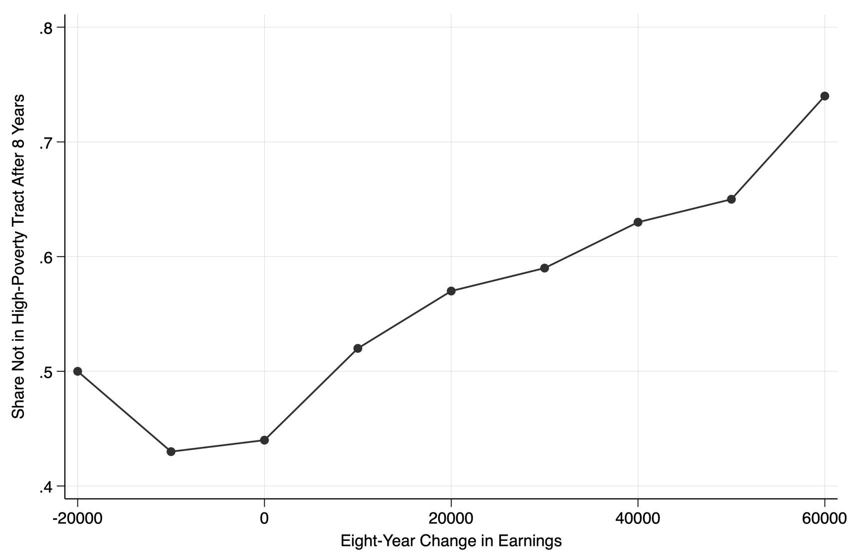
Source: Authors' calculations using data from the ACS and MAFARF.

Figure 4: Tract Mobility and Earnings Mobility Over Eight Years

(a) Movement to Higher-Income Tracts, by Baseline Tract Poverty



(b) Exit From High-Poverty Tracts

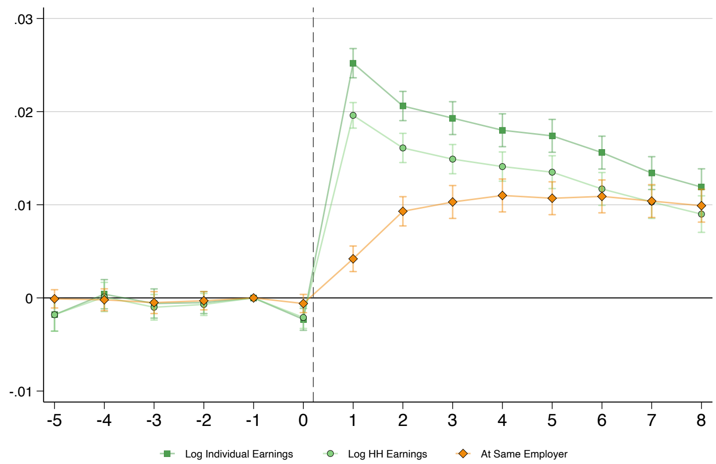


Notes: This figure is a binned scatter plot with the difference between the median income in an individual's baseline tract and their tract in $t+8$ on the Y-axis, and the eight-year change in the individuals' annual earnings on the X-axis. Individuals in Panel A are stratified based on the poverty rate of their baseline tract. Panel B shows probability of transitioning out of a high-poverty tract (baseline poverty rate over 30 percent) for those beginning in a high-poverty tract. The left- and right-most bins are open-ended. The figure uses the earnings sample, as described in Section 2.2, and changes are measured over the eight years following an individual's response to the ACS.

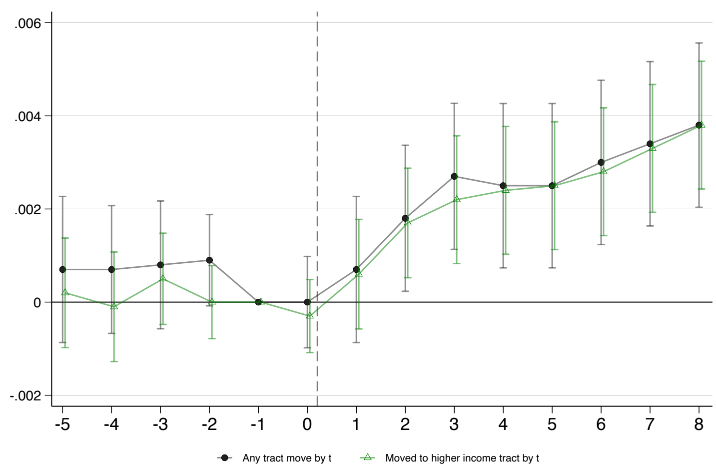
Source: Authors' calculations using data from the ACS, MAFARF, and LEHD.

Figure 5: Effects of a Coworker Earnings Shock on Own Earnings and Migration

(a) Effect of Shock on Own Earnings

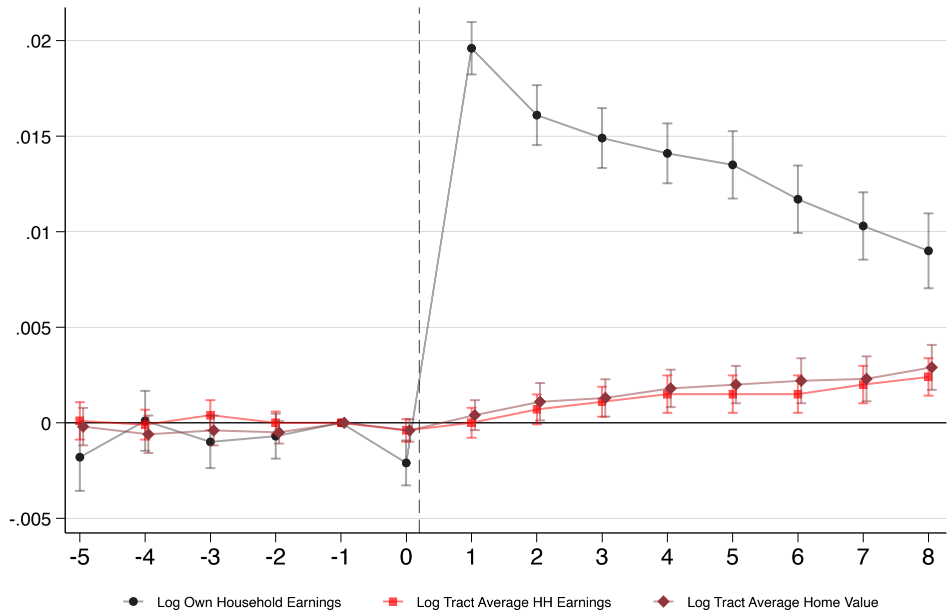


(b) Effect on Moving



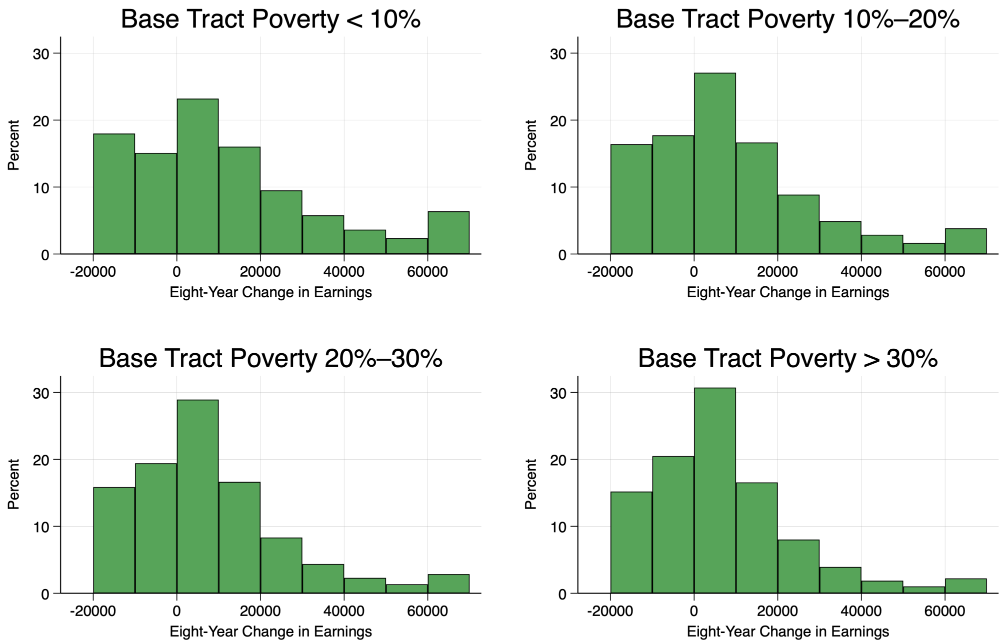
Notes: Figure displays OLS estimates of equation (1) with 95% confidence intervals. The figure uses a subset of the earnings sample, as described in Section 2.2. It includes individuals working at firms with at least 25 employees in the holdout sample, described in Section 4.2.2. Effects are scaled so that coefficients can be interpreted as the effect of a 10 percent change in coworker quarterly earnings between the base quarter in year t and the same quarter of $t + 1$. Panel A displays effects on own LEHD annual earnings, the household LEHD earnings (defined as the sum of ones own earnings and the earning of any other individual who was a spouse or co-householder in the base year), and an indicator for remaining at the same firm. Panel B displays reduced-form effects on the cumulative probability of having moved from one's baseline address by year t along with the unconditional cumulative probability of moving to a tract with higher average income (measure at baseline) than one's baseline tract. Standard errors are clustered at the firm level.

Figure 6: Effect of Earnings Shocks on Neighborhood Choice



Notes: Figure displays OLS estimates of equation (1) with 95% confidence intervals. The figure uses a subset of the earnings sample, as described in Section 2.2. It includes individuals working at firms with at least 25 employees in the holdout sample, described in Section 4.2.2. Effects are scaled so that coefficients can be interpreted as the effect of a 10 percent change in coworker quarterly earnings between the first quarter of period t and the first quarter of $t + 1$. Figure displays effects on the mean household earnings or median housing value of the individual's tract observed in the MAF in time t, where tract characteristics are held fixed using 2005–2009 ACS average values; we reproduce the effects on household earnings from Panel A of Figure 5 for comparison. Standard errors are clustered at the firm level.

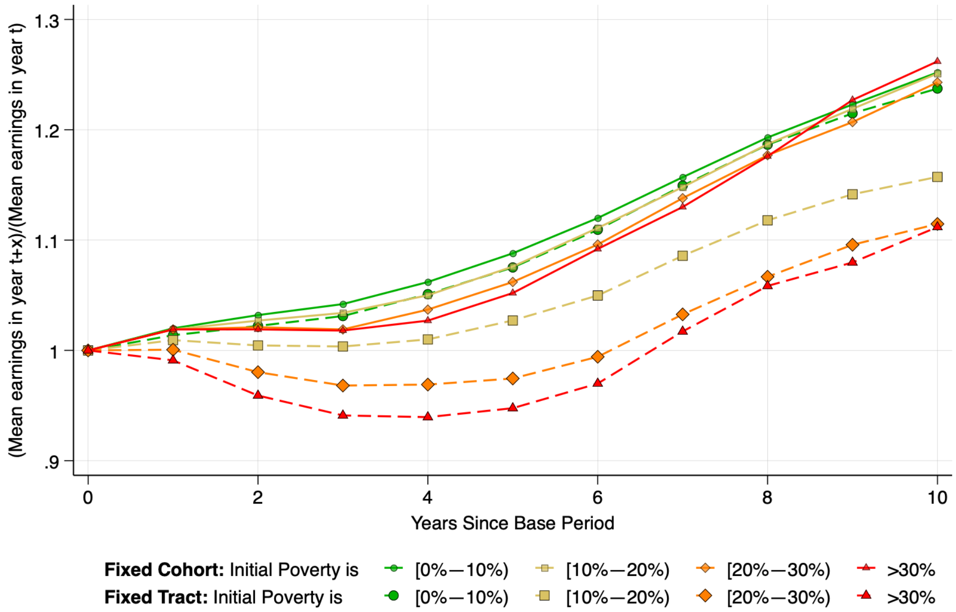
Figure 7: Distribution of Eight-Year Earnings Changes, By Baseline Tract Poverty Rate



Notes: This figure shows the distribution of eight-year earnings changes among people living in census tracts with different poverty rates in the baseline year. The left- and right-most bins are open-ended. The figure uses the earnings sample, as described in Section 2.2, and earnings growth is measured over the eight years following an individual's response to the ACS.

Source: Authors' calculations using data from the ACS, MAFARF, and LEHD.

Figure 8: Growth in Average Tract Earning Verus Earnings Growth of Baseline Residents



Notes: This figure compares the evolution of average earnings in neighborhoods at different poverty levels to the evolution of earnings among the *cohort* of people who lived in each neighborhood type in the baseline year. The contemporaneous earnings appear in the solid lines, and the cohort earnings are in the dashed lines. The exercise uses the earnings sample, as described in Section 2.2, and further restricts to the 1965 to 1980 birth cohorts to minimize the role of retirement and labor force entry. Mean earnings in each year are normalized by the value in the baseline year, and baseline years from 2005 to 2009 are included.

Source: Authors' calculations using data from the ACS, MAFARF, and LEHD.

Table 1: Eight-Year Transition Rates Across Neighborhood Types

Panel A: Neighborhood poverty rate

	Year $t+8$				
Year t	0–10	10–20	20–30	30+	Obs.
0–10	88.4	8.7	2.2	0.8	7,370,000
10–20	19.8	73.6	4.6	2	3,890,000
20–30	16.1	14.9	64.6	4.5	1,300,000
30+	12.7	13.9	9.8	63.5	672,000

Panel B: Quintile of neighborhood median income

	Year $t+8$					
Year t	1	2	3	4	5	Obs.
1	71.8	10.4	8.2	6.1	3.5	2,070,000
2	7.8	70.3	9.2	7.7	5	2,610,000
3	5.1	7.5	70.6	9.5	7.3	2,780,000
4	3.2	5.3	7.8	72.5	11.2	2,820,000
5	1.8	3.1	5.2	9.1	80.9	2,950,000

Notes: Table reports the share of people who start in each type of neighborhood in year t that end up in each type of neighborhood in year $t+8$, where t is the year they were sampled in the ACS. We classify census tracts based on 2005–2009 poverty rates in Panel A and population-weighted quintiles of the 2005–2009 median household income distribution in Panel B. This exercise uses the migration sample, as described in Section 2.2.
Source: Authors' calculations using data from the ACS and MAF.

Table 2: Initial Residential Neighborhood and Subsequent Earnings Dynamics

	(1)	(2)	(3)
Panel A: DV: Eight-year change in earnings, dollars			
Poverty rate 0–10	4143	3425	1147
	(136.3)	(132.3)	(129.5)
Poverty rate 10–20	2203	2015	473
	(144.2)	(138.8)	(130.4)
Poverty rate 20–30	851.5	694.1	-68.83
	(160.9)	(154)	(142.3)
Intercept	5726	6234	8110
	(125.1)	(123.5)	(121.8)
Observations	1,674,000	1,674,000	1,674,000
R-squared	0.0015	0.0185	0.059
CBSA and year FE		X	X
Individual controls			X
Panel B: DV: Eight-year change in earnings, arc percent			
Poverty rate 0–10	-0.0025	-0.0061	-0.006
	(0.0028)	(0.0028)	(0.0029)
Poverty rate 10–20	-0.0014	-0.0034	-0.0059
	(0.003)	(0.0029)	(0.0029)
Poverty rate 20–30	-0.0038	-0.0048	-0.0065
	(0.0034)	(0.0033)	(0.0032)
Intercept	0.104	0.1069	0.1076
	(0.0028)	(0.0028)	(0.0028)
Observations	1,674,000	1,674,000	1,674,000
R-squared	0	0.0135	0.0401
CBSA and year FE		X	X
Individual controls			X

Notes: Table reports regressions in which the dependent variable is the eight-year change in earnings experienced by an individual, measured in dollars (Panel A) or the arc percent (Panel B). The key explanatory variables of interest are indicators for the poverty rate of the neighborhood where an individual is observed at the start of the eight-year period; the omitted indicator is for the poverty rate above 30%. Columns 2 and 3 include CBSA and year fixed effects. Column 3 includes fixed effects for individual race, age, and baseline education. The exercise uses the earnings sample, as described in Section 2.2, and changes are measured over the years following an individual's response to the ACS. Standard errors, clustered at the level of the baseline tract, are reported in parentheses.

Source: Authors' calculations using data from the ACS, LEHD, and MAF.

Table 3: Exposure to High Poverty Neighborhoods

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Share by years in high poverty tract			Mean 5-year change in poverty rate,	Share spending 10 years in tract and poverty rate falls	Mean 2009–19 change in tract poverty rate, column 6
	Observations	≤5	6–9	10	movers	below 25%	sample
Overall	530,000	0.309	0.153	0.538	-0.237	0.127	-0.148
Age 25–35	92,000	0.502	0.19	0.309	-0.239	0.07	-0.14
Age 35–45	100,000	0.364	0.165	0.471	-0.235	0.117	-0.147
Age 45–55	120,000	0.264	0.145	0.59	-0.233	0.142	-0.148
Age 55–65	110,000	0.207	0.123	0.67	-0.238	0.163	-0.15
Over age 65	100,000	0.156	0.133	0.711	-0.246	0.153	-0.151
Black	180,000	0.29	0.173	0.537	-0.238	0.093	-0.144
Hispanic	130,000	0.27	0.141	0.589	-0.225	0.145	-0.151
White	190,000	0.353	0.138	0.509	-0.243	0.149	-0.148
Other	28,000	0.383	0.164	0.453	-0.247	0.134	-0.153
Owner-occupant	300,000	0.217	0.124	0.659	-0.245	0.173	-0.151
Renter	220,000	0.405	0.184	0.412	-0.233	0.079	-0.139
No kids	320,000	0.275	0.143	0.582	-0.241	0.137	-0.149
Has kids	210,000	0.352	0.166	0.482	-0.234	0.114	-0.146
No kids, under 40	42,000	0.537	0.175	0.288	-0.247	0.071	-0.147
No kids, over 40	280,000	0.223	0.137	0.641	-0.238	0.15	-0.149
Has kids, under 40	100,000	0.437	0.187	0.376	-0.235	0.088	-0.141
Has kids, over 40	110,000	0.262	0.143	0.595	-0.231	0.142	-0.149

Notes: Table provides summary statistics among individuals in the migration sample who are observed living in a high poverty neighborhood (where the 2005–2009 poverty rate exceeds 30%) in the year they are sampled in the ACS. Column 1 reports the number of observations. Columns 2–4 report the share of individuals who spend less than or equal to 5, 6 to 9, or 10 of the subsequent 10 years in a high-poverty neighborhood. Column 5 reports the mean 5-year change in the poverty rate among individuals who move out of a high-poverty neighborhood by year 5. Column 6 reports the share of individuals who spend 10 years in a tract that initially has a poverty rate above 30% but falls below 25% in the 2015-2019 ACS. Column 7 reports the mean 2009–2019 change in the poverty rate among the sample of stayers in column 6. We restrict to individuals who were observed in the MAFARF in at least 8 of the 10 years following their ACS year, proportionally adjusting exposure of those observed in fewer than 10 years.

Source: Authors' calculations using data from the ACS and MAF.

Online Appendix

Table A.1: One-Year Transition Rates Across Neighborhood Types

Panel A: Neighborhood poverty rate

	Year $t+1$			
Year t	0–10	10–20	20–30	30+
0–10	97.2	2	0.5	0.2
10–20	4.2	94	1.2	0.6
20–30	3.5	3.7	91.5	1.3
30+	2.7	3.4	2.7	91.2

Panel B: Quintile of neighborhood median income

	Year $t+1$				
Year t	1	2	3	4	5
1	93.5	2.6	1.9	1.3	0.7
2	2.1	93.1	2.1	1.7	1
3	1.3	1.8	93.3	2.1	1.4
4	0.8	1.2	1.8	94	2.2
5	0.4	0.7	1.1	1.9	96

Notes: Table reports the share of people who start in each type of neighborhood in year t that end up in each type of neighborhood in year $t+1$, where t is the year they were sampled in the ACS. We classify census tracts based on 2005–2009 poverty rates in Panel A and population-weighted quintiles of the 2005–2009 median household income distribution in Panel B. This exercise uses the migration sample, as described in Section 2.2.
Source: Authors' calculations using data from the ACS and MAF.